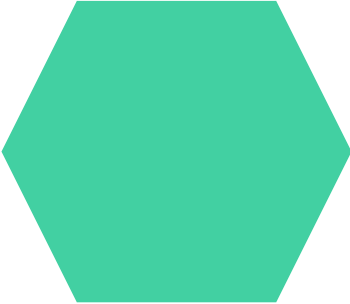




**PROJECT TITLE -**



**HEALTHCARE ANALYTICS  
FOR DOCTOR VISITS**

**NAME : UJJWAL GUPTA**

**AICTE STU ID :**

**STU68374882bc9891748453506**



# PROBLEM STATEMENT

- **Staffing Issues:** Hospitals struggle to predict how many patients will visit, causing long waiting lines and staff stress.
- **Different Patient Needs:** People of different ages, genders, and incomes use healthcare in completely different ways.
- **Insurance Gaps:** It is unclear how government medical aid for low-income and older patients affects clinic visits compared to private insurance.



# PROJECT DESCRIPTION



- **Smart Data Ingestion:** Analyzed a clean, real-world database of 5,190 patient records with no missing information.
- **Tracking Key Factors:** Studied how things like gender, income, and how sick a patient feels change how often they visit the doctor.
- **Better Health Logistics:** Uses data patterns to help hospital managers know exactly when they need more doctors or medicine on hand.



# WHO ARE THE END USERS?

- **Hospital Managers:** They use the data to schedule the right number of doctors and stop long checkout queues.
- **Health Policy Makers:** They can check if government free medical benefits are helping low-income and older citizens effectively.
- **Medical Researchers:** They study how an individual's background and temporary illnesses affect actual clinical visits.
- **Medicine Suppliers:** They use these patterns to send vaccines and medicines to the clinics that need them most.

# TECHNOLOGY USED

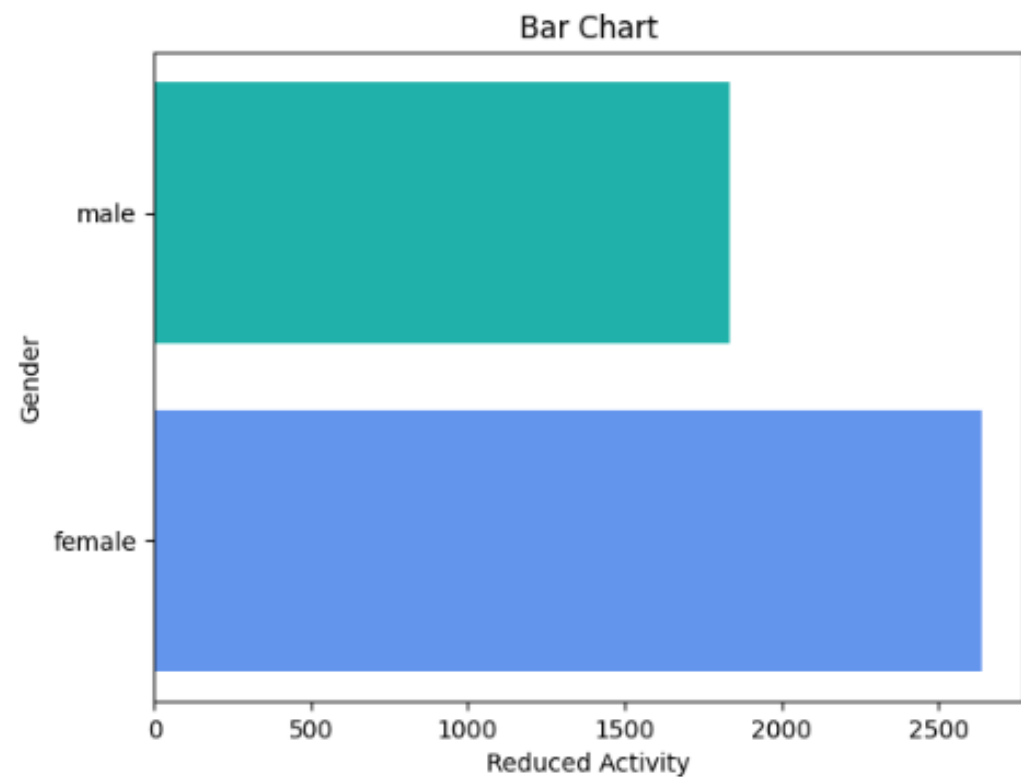
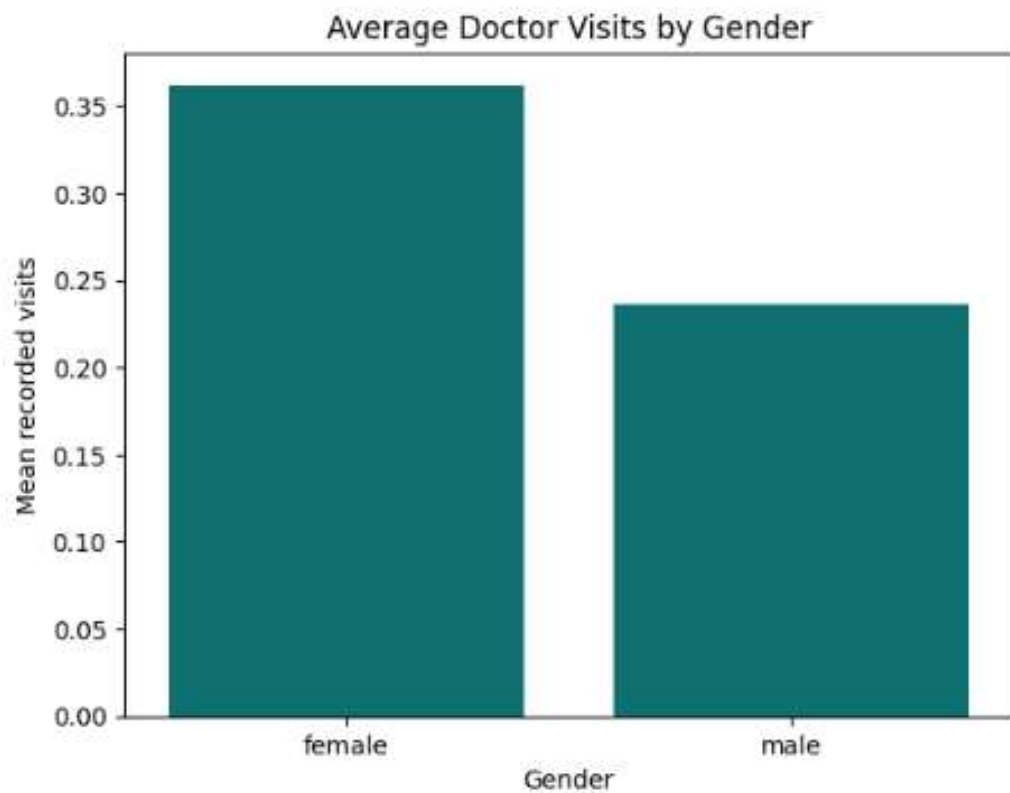
- **Python:** The core programming language used to load and calculate all statistical metrics.
- **Pandas:** Used to clean the dataset, handle empty fields, and sort rows and columns.
- **NumPy:** Used to perform fast mathematical calculations and process data arrays efficiently.
- **Matplotlib & Seaborn:** Used to build simple graphs, chart layouts, and color-coded comparison tables.
- **Google Colab:** The online notebook environment where all the data code was written and run.



# RESULTS

- **Top Driver:** Immediate sickness and restricted activity days are the main reasons people visit the doctor. 
- **Income Factor:** Lower-income individuals visit doctors much more frequently than higher-income groups.
- **Activity Impact:** Patients with 14 days of restricted activity go to the doctor **10 times more often** than those with 0 days.
- **Demographics:** The data skews slightly female (52%), with high reliance on public health aid programs.

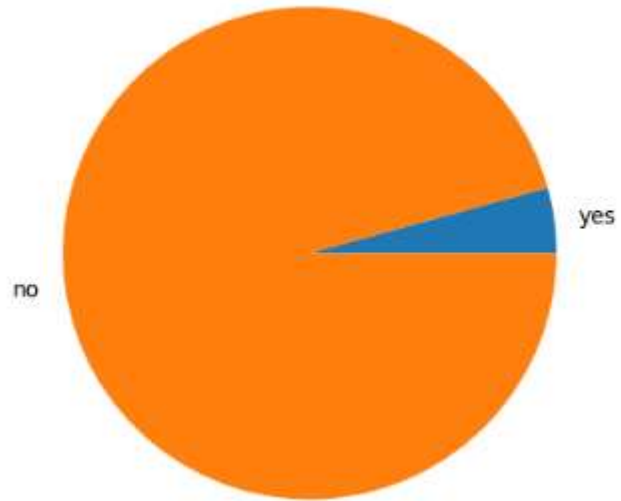
# RESULTS



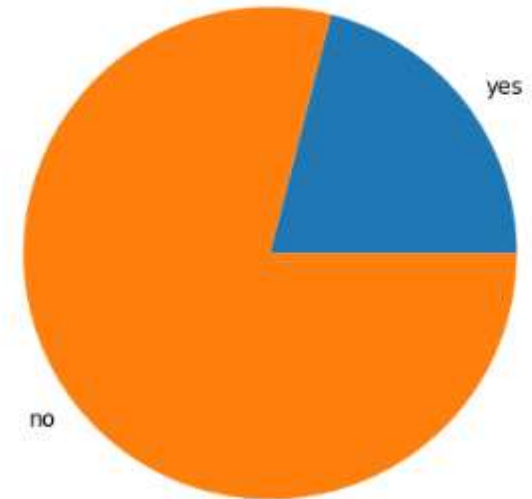
[Github Link](#)

# RESULTS

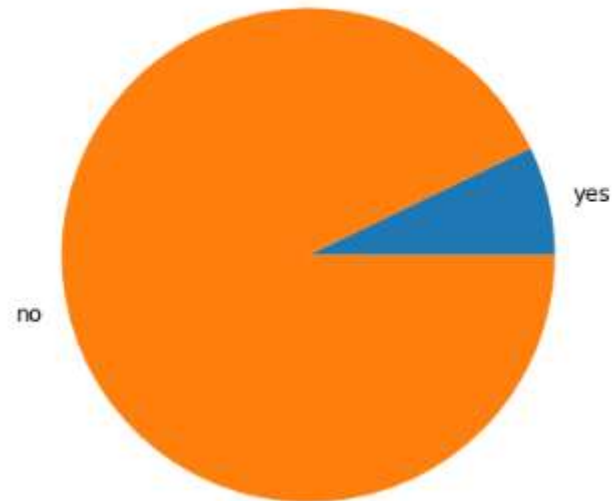
% of people getting Govt Health Insurance due to low income



% of people getting Govt Health Insurance due to old age, disability or veteran status

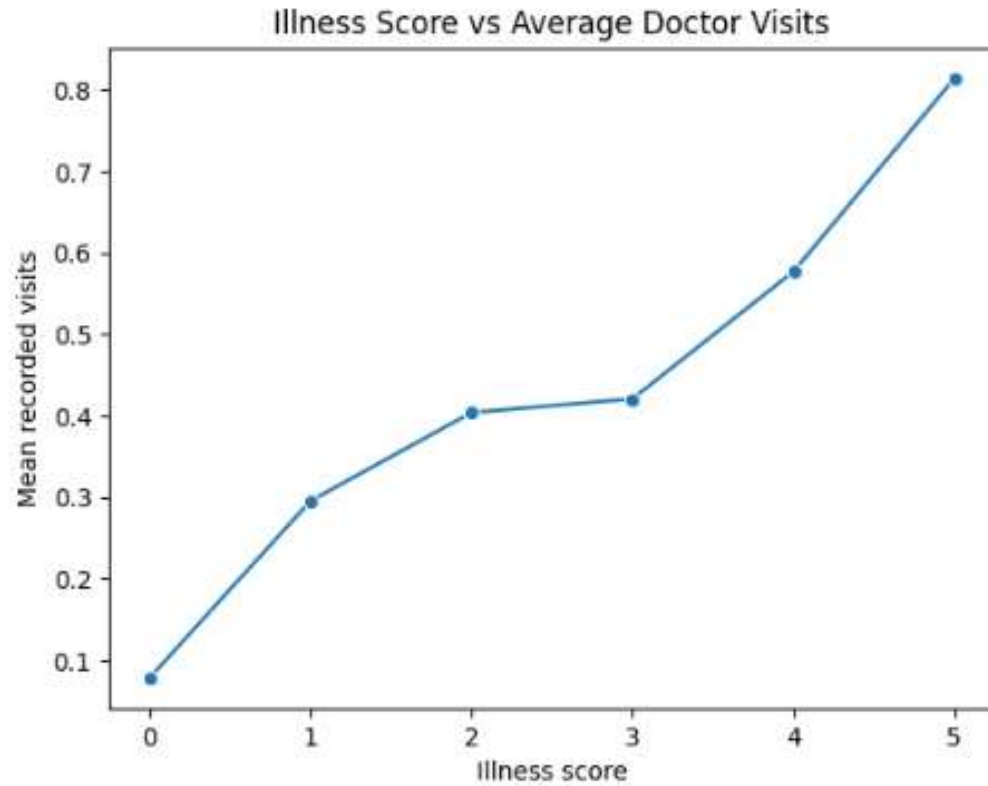


% of people having Private Health Insurance

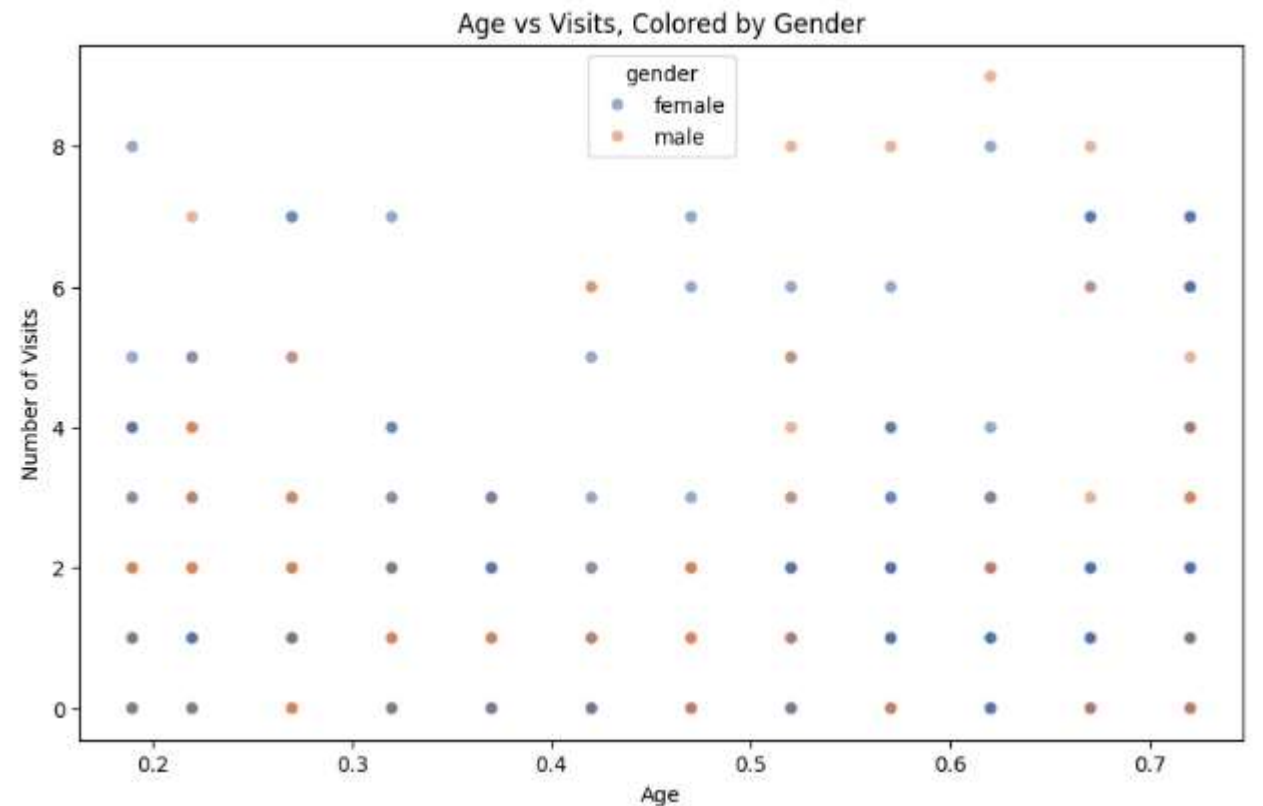


[Github Link](#)

# RESULTS



[Github Link](#)





# THANK YOU

